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In the leaves, the ground tissue consists of chlorenchyma tissues. This region is called mesophyll. In hydrophytes, cortex is **Aerenchymatous** (with air cavities).

Its general function is storage of food as well as providing mechanical support to organs.

Endodermis

The cells of this layer are barrel shaped and arranged compactly without intercellular spaces.

Endodermis is the innermost cortical layer that separates cortex from the stele. This layer may be a true endodermis as in root or it is an endodermis like layer in stems. This layer is morphologically homologous to the endodermis found in the root.

The cells of endodermis like layer had living cells containing starch grains. Hence it is known as starch sheath. In true root endodermis, radial and inner tangential walls of endodermal cells possess thickenings of **lignin, suberin and some other carbohydrates** in the form of strips they are called **casparian strips**.

The endodermal cells, which are opposite to the protoxylem elements, are thin walled without casparian strips. These cells are called **passage cells**. Their function is to transport water and dissolved salts from the cortex to the protoxylem.

Water cannot pass through other endodermal cells due to casparian strips. The main function of casparian strips in the endodermal cells is to prevent the re-entry of water into the cortex once water entered the xylem tissue.

The other suberized cells acts as water-tight layer between vascular and non-vascular regions to check the loss of water.

Pericycle

Pericycle is single or few layered parenchymatous found inner to the endodermis. It is the outermost layer of the stele. Rarely thick walled sclerenchymatous. In angiosperms, pericycle gives rise to lateral roots.

Pith or Medulla

The central part of the ground tissue is known as pith or medulla. Generally this is made up of thin walled parenchyma cells with intercellular spaces. The cells in the pith generally stores starch, fatty substances, tannins, phenols, calcium oxalate crystals, etc.

Albuminous Cells: The cytoplasmic nucleated parenchyma, is associated with the sieve cells of Gymnosperms. Albuminous cells in *Conifers* are analogous to companion cells of Angiosperms. It also called as strasburger cells.

9.6 Vascular Tissue System

This section deals with the vascular tissue system of gymnosperms and angiosperms stems and roots.The vascular tissue system consists of xylem and phloem. The elements of xylem and phloem are always organized in groups. They are called **vascular bundles.**

The stems of both groups have an eustele while roots are protostele. In eustelic organization, the stele contains usually a ring of vascular bundles separated by interfascicular region or medullary ray

The structural and organizational variation in vascular bundles is shown below.